

16/09/26

## Signal Processing

### Announcements

- Recap: Importance of phase, Eigen function

### Useful DTFT properties:

- $x^*[n]$ ,  $x^*[-n]$ ,  $\text{Re}\{x[n]\}$ ,  $j\text{Im}\{x[n]\}$ ,

$$x_e[n] = \frac{1}{2} (x[n] + x^*[-n]),$$

$$x_o[n] = \frac{1}{2} (x[n] - x^*[-n]),$$

Any real  $x[n]$ .

- DTFT Theorems: Linearity, delay by  $n_d$ , modulation by  $\cos$ , convolution

- Recap: •  $\text{DTFT}\{x^*[n]\} = X^*(e^{-j\omega})$  (1) (where  $\text{DTFT}\{x[n]\} = X(e^{j\omega})$ )

$$\begin{aligned} \text{DTFT}\{x^*[-n]\} &= \sum_{n=-\infty}^{\infty} x^*[-n] e^{-j\omega n} = \sum_{k=-\infty}^{\infty} x^*[k] e^{j\omega k} \\ &= \left( \sum_{k=-\infty}^{\infty} x[k] e^{-j\omega k} \right)^* \\ &= X^*(e^{j\omega}) \quad (2) \end{aligned}$$

- $\text{Re}\{x[n]\} = ?$

$$\text{Re}\{x[n]\} = \frac{1}{2} (x[n] + x^*[n])$$

$$\Rightarrow \text{DTFT}\{\text{Re}\{x[n]\}\} = \frac{1}{2} (X(e^{j\omega}) + X^*(e^{-j\omega}))$$

$$= X_e(e^{j\omega}) \quad (\text{even part of } X(e^{j\omega}))$$

Verify that  $X_e(e^{j\omega}) = X^*(e^{-j\omega})$

$$\text{DTFT}\{j\text{Im}\{x[n]\}\} = \text{DTFT}\left\{\frac{1}{2} (x[n] - x^*[n])\right\}$$

$$= \frac{1}{2} (X(e^{j\omega}) - X^*(e^{-j\omega})) = X_o(e^{j\omega}) \quad (\text{odd part of } X(e^{j\omega}))$$

$$\begin{aligned}
\bullet \text{ DTFT} \{x_e[n]\} &= \text{DTFT} \left\{ \frac{1}{2} (x[n] + x^*[-n]) \right\} \\
&= \frac{1}{2} (\text{DTFT} \{x[n] + x^*[-n]\}) \\
&= \frac{1}{2} \text{DTFT} \{x[n]\} + \text{DTFT} \{x^*[-n]\} \quad \left. \begin{array}{l} \text{linearity} \\ \text{of DTFT} \end{array} \right\} \\
&= \frac{1}{2} (X(e^{j\omega}) + X^*(e^{j\omega})) = \text{Re}\{X(e^{j\omega})\}
\end{aligned}$$

Recall:  $\text{DTFT} \{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$  (linearity follows from the properties of complex addition/multiplication)

$$\begin{aligned}
\bullet \text{ DTFT} \{x_o[n]\} &= \text{DTFT} \left\{ \frac{1}{2} (x[n] - x^*[-n]) \right\} \\
&= j \text{Im}\{X(e^{j\omega})\}
\end{aligned}$$

↓ Fill in the intermediate steps

• Let's look at real signals now. i.e.,

$$x[n] = x^*[n] \quad - (3)$$

$$\Rightarrow \text{DTFT}\{x[n]\} = \text{DTFT}\{x^*[n]\}$$

$$\text{i.e., } X(e^{j\omega}) = X^*(e^{-j\omega}) \quad - (4)$$

of  $X(e^{j\omega}) = X_R(e^{j\omega}) + j X_I(e^{j\omega})$ , (5) what can we say about the symmetry properties of  $X_R(e^{j\omega})$ ,  $X_I(e^{j\omega})$ ?

From (4) (5), we see that

$$X_R(e^{j\omega}) + j X_I(e^{j\omega}) = X_R(e^{-j\omega}) - j X_I(e^{-j\omega})$$

$$\Rightarrow X_R(e^{j\omega}) = X_R(e^{-j\omega}), \text{ i.e.; even symmetry } \quad (6)$$

$$X_I(e^{j\omega}) = -X_I(e^{-j\omega}), \text{ i.e., odd symmetry. } \textcircled{7}$$

- These results translate to  $|X(e^{j\omega})|$  being even, and  $\angle X(e^{j\omega})$  being odd function of  $\omega$ . This follows from the defn of  $|X(e^{j\omega})|$ ,  $\angle X(e^{j\omega})$

$$|X(e^{j\omega})| = \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})}$$

$$\angle X(e^{j\omega}) = \tan^{-1} \left( \frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right)$$

### • DTFT theorems:

- Linearity:  $\text{DTFT}\{a x[n] + b y[n]\} = a \cdot \text{DTFT}\{x[n]\} + b \cdot \text{DTFT}\{y[n]\}$

- Show this

- Delay:  $\text{DTFT}\{x[n-n_d]\} = \sum_{n=-\infty}^{\infty} x[n-n_d] e^{-j\omega n}$   $n_d$  is finite

$$= \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega(m+n_d)} \quad \text{where } m = n - n_d$$

$$= \underbrace{e^{-j\omega n_d}} \cdot X(e^{j\omega})$$

- Modulation:  $\text{DTFT}\{\underbrace{e^{j\omega_0 n}} x[n]\} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \cdot e^{j\omega_0 n}$   
 $= \sum_{n=-\infty}^{\infty} x[n] e^{-jn(\omega - \omega_0)}$   
 $= X(e^{j(\omega - \omega_0)})$